

OHC Master Mind

Dynamic Cone Loading & Unloading

Application guide and system description — plant setup, PLC data source, simulation and the central decision engine

16 · 4 · 11

loaders · unloaders · pallet stations

380

baskets on a 304 m OHC loop, 1 cone each

10

variants, 35 cones/h each
(Requirements.xlsx)

What this book covers. How the application is used screen by screen — every screenshot is marked with numbered arrows and explained — how the Master Mind brain makes its decisions, how the conveyor timing and basket positions are calculated, how the PLC is connected over Modbus TCP, the results of the simulation so far, and the questions that still need answers from the site.

Version 1.0 · 28 September 2026 · Status: simulation and PLC interface complete; live connection to the site PLC pending site visit.

Values marked “assumed” are not in the requirement sheet and must be confirmed.

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Introduction

The problem

In the OHC (overhead conveyor) cone-handling system, 16 autoconers wind cones and hand them to 16 loaders. The loaders put the cones — one per basket — onto a closed chain of 380 baskets. Four unloaders take the cones off and place them in the accumulators of 11 pallet stations, where pallets are built (4 or 5 cones per row, from the pallet matrix).

The customer specification uses a **FIFO queue**: when a loader reaches its count it joins the queue and loaders are served strictly in order, with at most 4 loaders loading at once (“4Q”) and at most 4 variants on the loop. With 10 variants this means some autoconers must wait, fill their buffer and stop, while baskets pass their unloader unloaded and go round again.

What we built

Part	What it does
Plant setup	Every entry of Requirements.xlsx, in the sheet’s order, with the customer values as defaults, plus the values the sheet does not give. It calculates all timings and checks the values before the plant is built.
PLC data source	Reads the live plant data from the PLC over Modbus TCP through a small gateway program, or runs a simulated PLC with random but consistent values. Checks the data against the setup and shows the information outputs.
Simulation	A digital twin of the plant built from the setup: every basket on the chain, loaders with their buffers, unloaders, pallet stations, RFID basket tracking. Runs FIFO (customer spec) or the Master Mind.
Master Mind	The central decision engine. Every second it decides which loader loads, how many cones, to which pallet station and at what pace, using timing forecasts. Every decision is explained on screen.
PLC gateway and simulator	Node.js programs: the gateway talks Modbus TCP to the PLC for the web page; the simulator behaves like the real PLC for testing without the plant.

Status

Simulation, decision engine, setup and PLC data interface are complete and tested (read and write over Modbus TCP against the PLC simulator). The next steps are the site visit, confirming the assumed values and the PLC data layout, and connecting the brain to the live PLC data.

338 /h
cones to pallets — Master Mind,
automatic variants (FIFO: 204 /h)

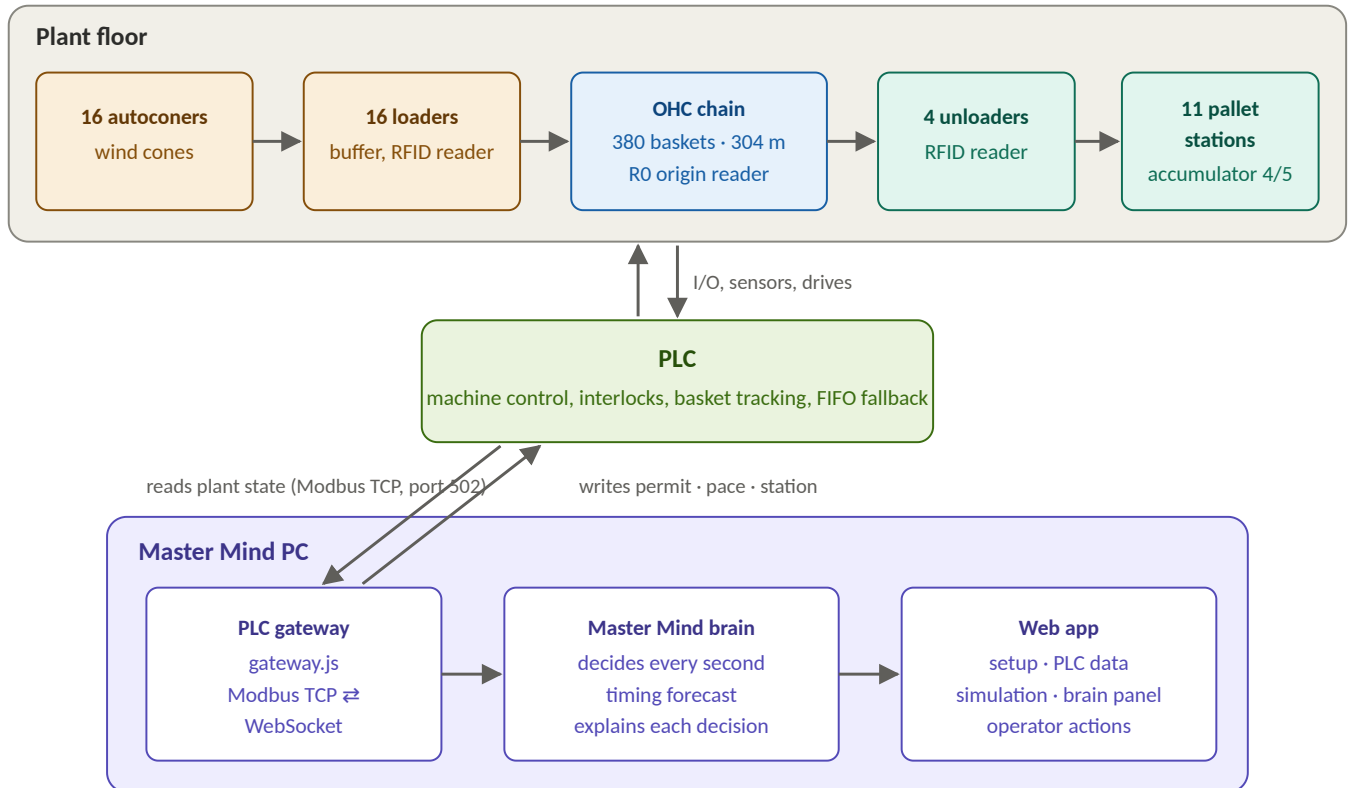
0 %
autoconer stop time (FIFO: 22 %)

16–42 min
to clear an old lot after a lot change (FIFO:
81–128 min)

Simulation results with the customer values and the assumed values listed in chapter 13 (4 simulated hours). They will be confirmed with the real values.

System architecture

The Master Mind runs on a separate PC next to the PLC. The PLC keeps doing all machine control and safety; the Master Mind only reads the plant state and writes decisions (which loader may load, at what pace, to which pallet station). If the Master Mind stops, the PLC continues with plain FIFO.



Data flow in one second

- Read.** The gateway reads all plant registers from the PLC (1,776 words in 18 requests, about 35 ms in the test).
- Decide.** The brain works out waiting times, buffer fill times, basket positions and unloader arrival times, and chooses what to do (chapter 9).
- Write.** Only when writing is allowed: `PERMIT_LOAD`, `LOAD_PACE` and `STATION` for each permitted loader, station assignments, and the brain heartbeat.
- Show.** The web app shows the plant, every decision with its reason, and the information outputs.

Safety. The Master Mind never controls safety functions (e-stop, guards, interlocks). The PLC watches the brain heartbeat; if it stops, the PLC falls back to FIFO with the same limits and production continues.

Plant data and timing

Inputs from Requirements.xlsx

#	Requirement	Value	#	Requirement	Value
1	Package / hour	380	11	Cones per hour per variant	35
2	No of variants	10	12	Loader vs variety	entry for all loaders
3	Loading time / package	20 s	13	No of unloaders	4
4	Unloading time / package	18 s	14	Lot change information	high priority
5	OHC loop length	(blank) → 304 m	15	Loading count	automatic / manual
6	Loop linear speed	10 m/min	16	Loader to start loading	demand / unloader
7	Distance to unloader	centre of the ring	17	Cones per accumulator	4 or 5
8	No of baskets	380	18	No of pallet stations	11
9	Pitch	800 mm	19	Information (dynamic)	3 live outputs
10	No of loaders	16			

Timing calculated from the inputs

Figure	Formula	Value
Loop length	baskets × pitch = 380×0.8 m	304 m
A basket passes a station every	pitch ÷ speed = $0.8 \text{ m} \div 0.167 \text{ m/s}$	4.8 s
Baskets passing a station per hour	$3600 \div 4.8$	750
One full loop	$304 \text{ m} \div 10 \text{ m/min}$	30 min 24 s
Loader cycle on the moving chain	20 s rounded up to whole baskets: $20 \div 4.8 = 4.2 \rightarrow$ every 5th basket	24 s · 150 /h
Unloader cycle on the moving chain	$18 \div 4.8 = 3.75 \rightarrow$ every 4th basket	19.2 s · 187 /h
Loading streams one unloader can take	$24 \div 19.2$	1.25
Loader → unloader travel	distance along the chain ÷ speed (L16→U01 ... L01→U04)	5 min 36 s – 17 min 36 s
Empty return	last unloader (176.8 m) → R0 (304 m)	12 min 43 s
Autoconer output	10 variants × 35 cones/h	350 cones/h

Why the station cycles are rounded up. The chain never stops. A loader that needs 20 s after a cone can only use a basket that is in front of it; baskets arrive every 4.8 s, so the 2nd, 3rd and 4th baskets pass while it is still busy and it uses the 5th. Its real cycle is $5 \times 4.8 = 24$ s. The same rule gives the unloader 19.2 s.

Station positions (measured from the origin R0)

Point	Rule	From R0	Pitch
R0 (RFID origin reader)	fixed point, 0 m = loop end	0 m	0
L01 ... L16	R0 → L01 0.8 m, then every 6.4 m (assumed)	0.8 ... 96.8 m	1 ... 121
U01	0.8 m + centre of ring (152 m)	152.8 m	191
U02 · U03 · U04	every 8 m (assumed)	160.8 · 168.8 · 176.8 m	201 · 211 · 221
Loop end	= R0	304 m	380

How many variants must be on the loop at once

A variant occupies the loop from the moment its loader starts until its last cone is unloaded. With the average travel of about 12.8 min, one batch holds the loop for about 19 min. A variant with two loaders needs about 0.37 of the loop time on average, a variant with one loader about 0.74. For the 10 variants this adds up to about **5.2 variants on the loop at once**. With a fixed limit of 4, some autoconers must wait and stop — which is why the Master Mind can decide the number of variants itself (automatic mode, chapter 9).

Installing and running the app

Requirements

- Windows or Linux PC with **Node.js 20 or newer** and a web browser (Chrome or Edge).
- Project folder: `sieger_production`.
- For the real PLC: network access from this PC to the PLC (Modbus TCP, normally port 502).

Commands (run in the project folder)

Command	What it does
<code>npm install</code>	Once: installs the app's packages.
<code>npm run dev</code>	Starts the app. Open the address it prints (for example <code>http://localhost:5173</code>).
<code>npm run build</code> · <code>npm run preview</code>	Builds the finished app into <code>dist/</code> and serves it.
<code>npm run plc:install</code>	Once: installs the PLC gateway packages (Modbus and WebSocket).
<code>npm run plc:gateway</code>	Starts the PLC gateway on <code>ws://localhost:8765</code> . Needed for “Actual PLC data”.
<code>npm run plc:simulator</code>	Starts a Modbus TCP PLC simulator on port 5020, for testing without the real PLC.

Folder structure

Path	Contents
<code>src/App.jsx</code>	Switches between plant setup, PLC data source, simulation and classic demo.
<code>src/dynamic/config.js</code>	Plant values, defaults, all timing and position calculations, checks.
<code>src/dynamic/dynamicBrain.js</code>	The Master Mind decision engine.
<code>src/dynamic/dynamicEngine.js</code>	The plant model: chain, loaders, unloaders, stations, RFID tracking, timing forecast, FIFO.
<code>src/dynamic/SetupPage.jsx</code> , <code>PlcPage.jsx</code> , <code>DynamicSim.jsx</code> , <code>DynamicPlant.jsx</code> , <code>DynamicBrainPanel.jsx</code>	The screens.
<code>src/plc/</code>	Modbus register map, simulated PLC, data checks, data source.
<code>plc-gateway/</code>	<code>gateway.js</code> (PLC ⇌ app), <code>plc-simulator.js</code> (test PLC), README.
<code>MASTER_MIND.md</code>	Technical notes.

The app remembers the last setup and PLC settings in the browser. **Customer defaults** on the setup page puts every value back to the requirement sheet.

Screen guide — Plant setup

The app opens on the plant setup. Every value the plant and the brain need is entered here; the defaults are the customer values from Requirements.xlsx.

5.1 Top of the setup page

1 · Plant setup | **2 · PLC data source**

Requirements.xlsx — plant entries

#	REQUIREMENT	ENTRY
1	Package / hour Sheet: 380	380
2	No of variant Sheet: 10	10
3	Loading time / package Sheet: 20 s	20
4	Unloading time / package Sheet: 18 s	18
5	OHC loop length Sheet: (blank)	0
6	Loop linear speed Sheet: 10 m/min	10
7	Distance to unloader Sheet: Centre of Ball ring	0
8	No of baskets Sheet: 380	380
9	Pitch Sheet: 800 mm	800
10	No of loaders Sheet: 16	16
11	Cones per hour per variant Sheet: 35	35
12	Entry provision: loader vs variety (for all loaders) Sheet: Empty	16 loaders assigned → 34s SCHV x2, 40s Compact x2, 30s Caded x2, 26s Slub x2, 60s Combed x2, 20s OpenEnd x2, 50s Modal x1, 80s Micro x1, 32s PC x1, 40s CVC x1 Rate per loader in "Loader → variant" below
13	No of unloaders Sheet: 4	4

What these values give

- OHC loop length: 304.8 m (380 basket positions)
- A basket passes a station every: 4.8 s
- One full loop: 36m 24s
- Distance to first unloader: 152.8 m · 15m 12s
- Autocorner output: 358 cones/h
- Design throughput: 380 packages/h
- Basket passes per hour: 750
- Loader cycle on the moving chain: every 5th basket = 24.8 s
- Unloader cycle on the moving chain: every 4th basket = 19.2 s
- Loader capacity: 150 /h each · 2400 /h all
- Unloader capacity: 187 /h each · 750 /h all
- Loading streams one unloader: 1.25 /24.8 + 19.2 s/
- can take
- Loader → unloader travel: 5m 36s → 17m 36s
- Empty return (last unloader → RO): 12m 43s
- Variety slots needed (estimate): 5.2 of auto x 11 (stations)
- Pallet stations per unloader: 3 · 3 · 3 · 2
- Loader start rule: demand + unloader
- Lot change: high priority

Buttons: Classic demo, Customer defaults, Build plant & start

Values are consistent — ready to build.

POSITIONS MEASURED FROM ORIGIN RO

POINT	FROM RO	PITCH
RO origin reader	0.0 m	0
L01	0.8 m	1
L02	7.2 m	9

Figure 5.1 — Setup page: tabs, requirement rows and the calculated values

- 1 Setup tabs.** “1 · Plant setup” is this page; “2 · PLC data source” connects the PLC (chapter 6). Values entered here are used on both tabs.
- 2 Requirement rows.** Numbered 1–19 exactly as in Requirements.xlsx. Under each requirement the sheet’s value is shown (“Sheet: 380”); the entry box on the right holds the value used.
- 3 Automatic values.** OHC loop length is blank in the sheet; 0 means “automatic” = baskets × pitch = 304 m.
- 4 Distance to unloader.** 0 means “centre of the ring”, as in the sheet (152 m from L01).
- 5 Classic demo.** The first, fixed 280-basket demo, kept for reference.
- 6 Customer defaults.** Puts every value back to the sheet values.
- 7 Build plant & start.** Builds the conveyor, loaders, unloaders, stations and the brain from these values and opens the simulation. Disabled while a value is out of range or an error is shown.
- 8 What these values give.** Live calculation of loop length, timing, capacities and positions (figures 5.5 and 5.6).

5.2 Requirement rows 10–19

The screenshot displays the 'Requirement rows 10–19' section of the SIEGER OHC Master Mind application. The interface is dark-themed. On the left, a list of requirements is shown with their respective sheet numbers. On the right, the configuration details for each requirement are displayed. Numbered callouts (1-7) point to specific settings:

- 1 Cones/h per variant**: Applies to all; change single variants in the variant table. (Value: 35)
- 2 Loader vs variety**: Summary; edit per loader in "Loader → variant".
- 3 Lot change priority**: High = old-lot cones loaded first, own pallet. (Value: High priority – old-lot cones loaded first, own pallet)
- 4 Loading count**: Automatic (brain decides) or Manual count. (Value: Automatic – software decides the trigger)
- 5 Loader to start loading**: Demand, unloader requirement, or both. (Value: Demand + unloader requirement)
- 6 Cones per accumulator**: 4 or 5 from the pallet matrix. (Value: 5 cones (matrix))
- 7 Information (dynamic)**: Shown live on the simulation and PLC pages. (Value: Category-wise packages in OHC conveyor, No of packages returned after unloader, Lot change remaining cones)

Figure 5.2 — Rows for cones/h, loader entry, lot change, loading count, start rule, accumulator and information

- 1 Cones per hour per variant.** Sets all variants at once; different values per variant are entered in the variant table.
- 2 Loader vs variety.** A summary of which variant each loader makes; each loader is set in the "Loader → variant" grid below.
- 3 Lot change priority.** High: after a lot change the old-lot cones are loaded first and keep their own pallet. Normal: own pallet, no extra priority.
- 4 Loading count.** Automatic: the brain decides how many cones a loader collects before it asks to load. Manual: a fixed count.
- 5 Loader to start loading.** Choose on demand (autoconer side), on unloader requirement, or both.
- 6 Cones per accumulator.** 4 or 5 from the pallet matrix — one accumulator row goes onto the pallet at a time.
- 7 Information (dynamic).** The three live outputs of the sheet: packages in the OHC by category, packages returned after the unloader, lot change remaining cones.

5.3 Variant table and loader entry

Variant table (rows 2, 11, 14, 16)

Variant name	Cones / h	Loaders	Current lot no.	High demand
34s SCHY	35	2	1	*
40s Compact	35	2	1	*
30s Carded	35	2	1	*
24s Slub	35	2	1	*
60s Combed	35	2	1	*
20s OpenEnd	35	2	1	*
50s Modal	35	1	1	*
80s Micro	35	1	1	*
32s PC	35	1	1	*
40s CVC	35	1	1	*

Loader → variant (row 12)

Loader	Variant
L01	34s SCHY
L02	40s Compact
L03	30s Carded
L04	24s Slub
L05	60s Combed
L06	20s OpenEnd
L07	50s Modal
L08	80s Micro
L09	32s PC
L10	40s CVC
L11	34s SCHY
L12	40s Compact
L13	30s Carded
L14	24s Slub
L15	60s Combed
L16	20s OpenEnd

Callouts:

- Variant name:** Type your real material names
- Cones / h:** Output of this variant; it is shared between the loaders that make it.
- Loaders:** How many loaders make this variant
- Current lot no.:** Starting lot number of the variant
- High demand:** Priority when the start rule includes demand
- Loader → variant:** Row 12: one entry for every loader

Figure 5.3 — Variant table (rows 2, 11, 14, 16) and the loader → variant grid (row 12)

- Variant name.** Type the real material names (for example “34s SCHY”); they appear everywhere in the app.
- Cones / h.** Output of this variant; it is shared between the loaders that make it.
- Loaders.** How many loaders are set to this variant (“none” in amber if no loader makes it).
- Current lot no.** The lot each variant starts with. The lot change itself is done live in the simulation.
- High demand.** Gives the variant priority when the start rule includes demand; can also be switched live with ★.
- Loader → variant.** One drop-down per loader — the “entry provision loader vs variety for all loaders” of row 12.

5.4 Other settings (not in the sheet)

Other settings (not in the sheet — assumed, change to the real values)

- Station positions:** From R0 → L01, spacings and d. (0.8 m)
- Origin R0 → L01:** RFID reference reader → L01 (0.8 m)
- Loader buffer:** Cones before the autoconer stops (assumed 20)
- Cones per pallet:** Assumed 100
- Max loading at once:** The customer’s “4Q”
- Fixed variant limit:** Spec value 4; FIFO always uses it
- Variants on the loop:** Automatic (brain) or Fixed
- Loading / unloading:** On the move, or chain stops

Figure 5.4 — Settings the requirement sheet does not give; defaults are assumptions

- Station positions.** Calculated from R0 → L01, the spacings and the distance to the unloader — or “From the layout”: type each loader and unloader position in metres from R0.
- Origin R0 → L01.** Distance from the RFID origin reader to the first loader (0.8 m = one basket).
- Loader buffer.** Cones a loader can hold before its autoconer must stop (assumed 20).
- Cones per pallet.** When a pallet is full and dispatched (assumed 100).
- Loading / unloading.** “On the move” (chain never stops, default) or “chain stops while a station works”.
- Max loaders loading at once.** The customer’s “4Q”.
- Variants on the loop (brain).** Automatic — the brain decides from free pallet stations and unloader time — or Fixed.
- Fixed variant limit.** The specification’s 4; FIFO always uses it, the brain only in Fixed mode.

Also here: spacing between loaders (assumed 6.4 m) and between unloaders (assumed 8 m).

5.5 Calculated values



Figure 5.5 — “What these values give”: timing and capacity from the entries

- 1 **Loop length** 304 m = 380 × 0.8 m, with the number of basket positions.
- 2 **Basket every 4.8 s** = pitch ÷ speed.
- 3 **Loader cycle 24 s** — 20 s rounded up to whole baskets (every 5th basket).
- 4 **Unloader cycle 19.2 s** — 18 s rounded up (every 4th basket).
- 5 **Streams per unloader** 1.25 = 24 ÷ 19.2: one unloader can take one loader’s full-speed stream, not two.
- 6 **Travel time** from the nearest to the farthest loader–unloader pair.
- 7 **Variants needed** — estimate of variants that must be on the loop at once (5.2), against the limit.
- 8 **Warnings / errors.** Checked before building: baskets not fitting, overlapping stations, too few variant slots, capacity below output.

5.6 Positions from the origin R0

POSITIONS MEASURED FROM ORIGIN R0 0 m = 304.0		
POINT	FROM R0	PITCH
R0 origin reader	0.0 m	0
L01	0.8 m	1
L02	7.2 m	9
L03	13.6 m	17
L04	20.0 m	25
L05	26.4 m	33
L06	32.8 m	41
L07	39.2 m	49
L08	45.6 m	57
L09	52.0 m	65
L10	58.4 m	73
L11	64.8 m	81
L12	71.2 m	89
L13	77.6 m	97
L14	84.0 m	105
L15	90.4 m	113
L16	96.8 m	121
U01	152.8 m	191
U02	160.8 m	201
U03	168.8 m	211
U04	176.8 m	221
Loop end (= R0)	304.0 m	380

Every basket carries an RFID tag. R0 is the origin: at commissioning one loop is run and R0 records the order of all tags (B001, B002 ...). Then position = (places ahead of the basket (last read at R0 + pitch pulses since) × pitch, and the distance to a station = (station - basket) mod loop length. To enter the real positions from the layout drawing, set "Station positions" to "From the layout" under Other settings.
Loading / unloading time is the station cycle time when the chain runs.

Figure 5.6 — Every station measured from R0

- 1 Positions from R0.** Every loader and unloader in metres and basket pitches from the origin.
- 2 R0 = 0 m.** The RFID origin reader.
- 3 U01 at 152.8 m** = 0.8 m (R0 → L01) + 152 m (centre of the ring).
- 4 Loop end = R0** at 304 m; positions wrap back to 0.
- 5 How positions are found** from the RFID order list and pitch pulses (chapter 10).

Screen guide — PLC data source

Tab 2 of the setup decides where the live data comes from: the real PLC, or a simulated PLC. Both give exactly the same registers, so everything can be tested before the plant is connected.

6.1 Simulation mode

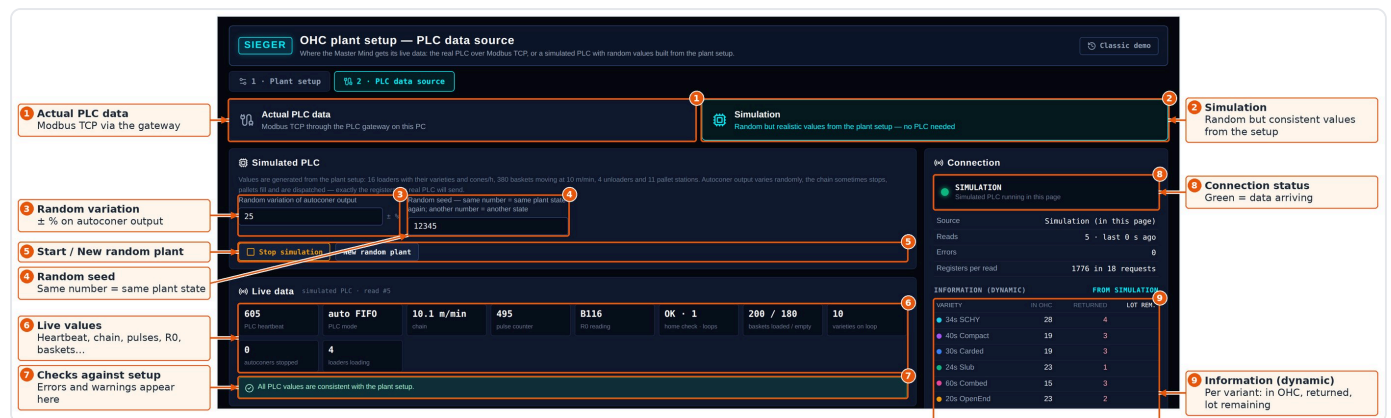


Figure 6.1 — Simulated PLC running in the page

- 1 **Actual PLC data.** Live data over Modbus TCP through the gateway (figure 6.3).
- 2 **Simulation.** A simulated PLC in the page: random but consistent values built from the setup (loaders and their varieties, 380 baskets at 10 m/min, unloaders, pallet stations).
- 3 **Random variation.** $\pm$ % variation of autoconer output (0 % = exactly the cones/h of the setup).
- 4 **Random seed.** The same number gives the same plant state again; another number gives another state.
- 5 **Start / New random plant.** Starts or stops the simulated PLC; “New random plant” restarts it with a new seed.
- 6 **Live values.** PLC heartbeat, mode, chain speed, pulse counter, basket at R0, home check, baskets loaded/empty, variants on the loop, stops, loaders loading.
- 7 **Checks against setup.** Wrong variety on a loader, more cones than the buffer, speed off by more than 10 %, heartbeat frozen, home-check error, faults.
- 8 **Connection status.** Green = data arriving; with reads, errors and registers per read.
- 9 **Information (dynamic).** From the PLC data: packages in the OHC per variant, returned after the unloader, lot change remaining.

6.2 Data tables

Figure 6.2 — Loader table (also unloaders, pallet stations, baskets and raw registers)

- Data views.** Loaders, unloaders, pallet stations, baskets (with position from R0) and the raw register values.
- Variety from the PLC**, marked "≠ setup" if it differs from the loader entry.
- Cones waiting** against the buffer; the bar turns amber and red as it fills.
- Loader status:** idle, ready, loading, fault, manual, off.
- Basket in front**, read by the loader's RFID reader.

6.3 Actual PLC (Modbus TCP)

1 PLC IP address.

2 Port — 502 normally; 5020 = test simulator

3 PLC IP address

4 Poll time — how often all registers are read (default 1000 ms).

5 Register type — holding registers (FC 03, 4xxxx) or input registers (FC 04, 3xxxx).

6 Gateway address — the gateway program on this PC (ws://localhost:8765).

7 Allow the brain to write. Commands are written to the PLC only when ticked; the gateway refuses writes otherwise.

8 Connect / Disconnect. The page reconnects by itself if the link drops.

9 PLC ONLINE — status with reads, errors and read time. Clear messages if the gateway is not running or the PLC cannot be reached.

10 Live values from the PLC — the same tiles, checks and tables as in simulation mode.

10 Actual PLC data
Modbus TCP through the PLC gateway on this PC

Simulation
Random but realistic values from the plant setup — no PLC needed

PLC connection (Modbus TCP)

PLC IP address: 127.0.0.1

Port: 5020

Unit ID (slave ID): 1

Poll time: 1000

Register type: Holding registers (FC 03, 4xxxx)

Gateway address (on this PC): ws://localhost:8765

Allow the brain to write commands to the PLC: ☐

Connect / Disconnect:

PLC ONLINE
Reading 127.0.0.1:5020

Source: Modbus TCP 127.0.0.1:5020 · unit 1

Errors: 5 · last 0 s ago

Reads: 1776 in 18 requests

Errors: 0

INFORMATION (DYNAMIC)

VARIETY	IN OHC	RETURNED	LOT #HW
34s SCHY	24	9	
40s Compact	26	13	
30s Carded	17	9	
24s Slub	20	11	
60s Combed	12	5	
20s OpenEnd	7	5	
50s Modal	5	5	

Live data

1563 auto FIFO 10 m/min 653 B274 OK · 1 168 / 212 10

PLC reads: 0

PLC mode: 4

Autocorner stopped: 0

Baskets loading: 4

Figure 6.3 — Connected to a PLC over Modbus TCP (here the included PLC simulator)

- PLC IP address.**
- Port** — 502 normally; 5020 for the included test simulator.
- Unit ID (Modbus slave ID).**
- Poll time** — how often all registers are read (default 1000 ms).
- Register type** — holding registers (FC 03, 4xxxx) or input registers (FC 04, 3xxxx).
- Gateway address** — the gateway program on this PC (ws://localhost:8765).
- Allow the brain to write.** Commands are written to the PLC only when ticked; the gateway refuses writes otherwise.
- Connect / Disconnect.** The page reconnects by itself if the link drops.
- PLC ONLINE** — status with reads, errors and read time. Clear messages if the gateway is not running or the PLC cannot be reached.
- Live values from the PLC** — the same tiles, checks and tables as in simulation mode.

6.4 Modbus register map

1 Block start addresses
Change to match the PLC program

2 Read blocks
PLC → brain: system, loaders, unloaders, stations, baskets

3 Command blocks
brain → PLC: permit, pace, station

4 Every register
Tag, address, description

5 CSV for the PLC programmer

6 Setup JSON
Run the Modbus PLC simulator with this plant

BLOCK	DIRECTION	ADDRESS RANGE	WORDS	PER ITEM
system	PLC → brain	0-15 (40001-40016)	16	1 × 16: plcHeartbeat, plcMode, chainRunning, speedX10, pulsesH, pulsesLo, r0Basket, encoderX1000, faultWord, homeCheck, loops, spare11, spare12, spare13, spare14, spare15
loaders	PLC → brain	100-227 (40101-40228)	128	16 × 8: cones, oldLoCones, status, stopped, variety, lot, loadedCount, basketInFront
unloaders	PLC → brain	300-323 (40301-40324)	24	4 × 6: status, basketInFront, unloadedCount, cycleX10, spare4, spare5
stations	PLC → brain	400-487 (40401-40488)	88	11 × 8: status, variety, lot, acc, pallet, palletPresent, palletsDone, spare7
baskets	PLC → brain	1000-2519 (41001-42520)	1520	380 × 4: status, variety, lot, target
cmdSystem	brain → PLC	3000-3001 (43001-43002)	2	1 × 2: brainHeartbeat, brainActive
cmdLoaders	brain → PLC	3010-3073 (43011-43074)	64	16 × 4: permit, pace, station, trigger
cmdStations	brain → PLC	3074-3117 (43075-43118)	44	11 × 4: variety, lot, command, spare

↓ Register map (CSV) ↓ Setup for the PLC simulator (JSON)

Figure 6.4 — Register map built from the plant setup

- 1 Block start addresses.** Change to match the PLC program; overlaps are flagged.
- 2 Read blocks** (PLC → brain): system, loaders, unloaders, pallet stations, baskets.
- 3 Command blocks** (brain → PLC): brain heartbeat, permit, pace, station, station assignment.
- 4 Every register** with tag, address and description.
- 5 CSV for the PLC programmer** — all 1,838 registers with 4xxxx addresses.
- 6 Setup JSON** — run the Modbus PLC simulator with exactly this plant.

Screen guide — Simulation

Build plant & start opens the simulation: a digital twin of the plant built from the setup, running either FIFO (customer specification) or the Master Mind.

7.1 Overview

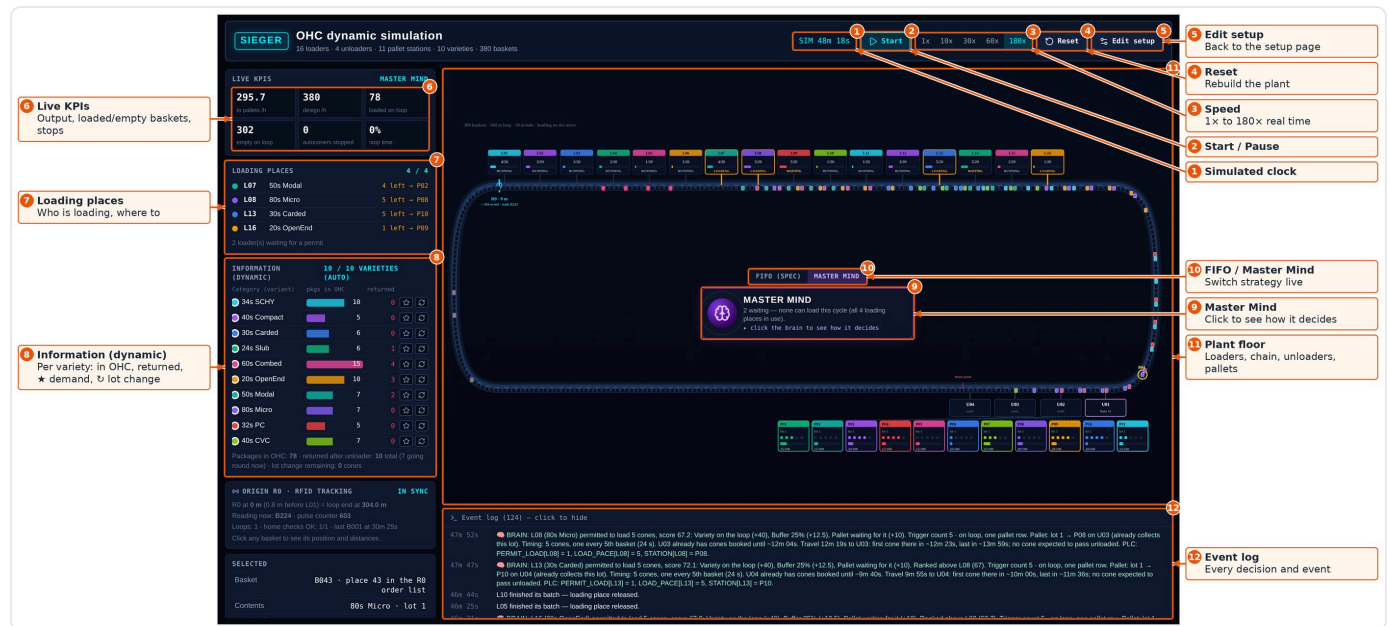


Figure 7.1 — Simulation page

- 1 **Simulated clock.** Time since the plant was built.
- 2 **Start / Pause.**
- 3 **Speed** — 1× (real time) to 180×.
- 4 **Reset** — rebuild the plant from the start.
- 5 **Edit setup** — back to the setup page.
- 6 **Live KPIs** — cones to pallets per hour, design rate, loaded and empty baskets, autoconers stopped, stop time.
- 7 **Loading places** — which loaders are loading, cones left and target pallet station.
- 8 **Information (dynamic)** — per variant: packages in the OHC, returned after the unloader, ★ high demand, ↻ lot change (with cones remaining of the old lot).
- 9 **Master Mind** — its current state; click it to open the decision panel (chapter 8).
- 10 **FIFO / Master Mind** — switch the strategy while running.
- 11 **Plant floor** — figure 7.2.
- 12 **Event log** — every decision and event with its time; brain decisions include the full reason and the PLC commands.

7.2 Plant floor

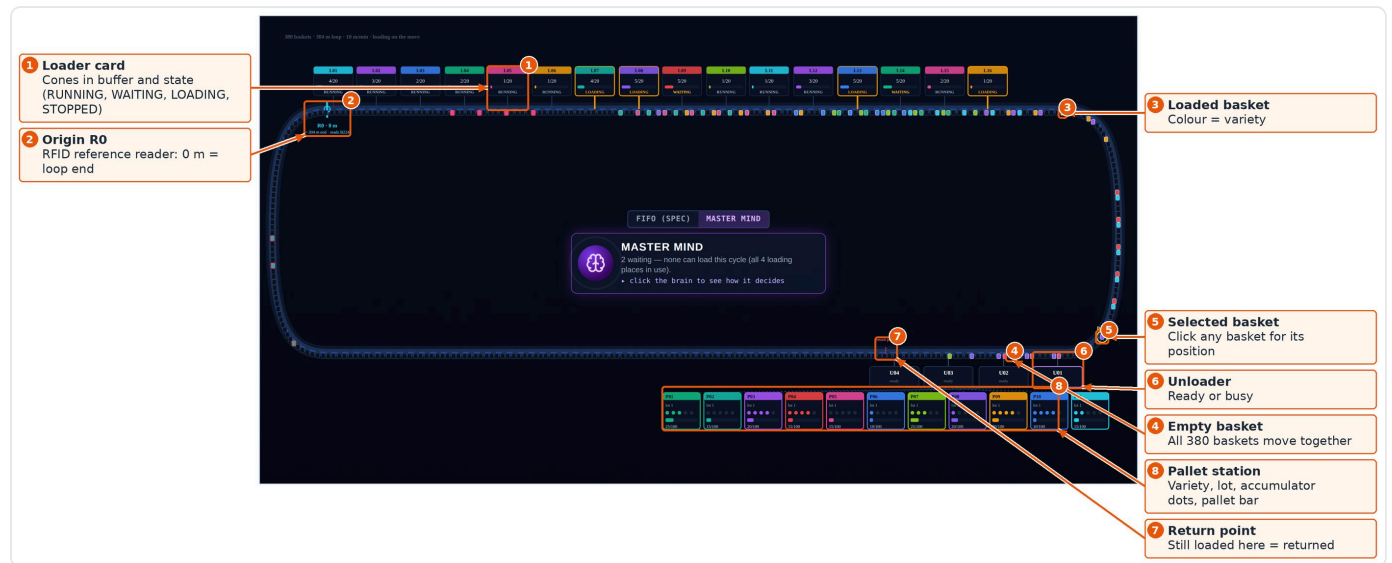


Figure 7.2 — Loaders, chain, unloaders and pallet stations

- 1 Loader card** — cones in the buffer (for example 4/20), bar, and state: RUNNING, FILLING, WAITING, LOADING or STOPPED (buffer full).
- 2 Origin R0** — the RFID reference reader: 0 m and loop end; shows the basket it is reading.
- 3 Loaded basket** — colour = variety. A red outline means it passed all unloaders and is going round again.
- 4 Empty basket** — all 380 baskets are drawn and move together, always in sequence.
- 5 Selected basket** — click any basket; its position and distances appear in the side panel (figure 7.3).
- 6 Unloader** — ready or busy (seconds left).
- 7 Return point** — after the last unloader; a basket still loaded here is counted as returned.
- 8 Pallet station** — variety and lot, accumulator dots (4 or 5), pallet bar and count; dashed = held (partial pallet kept).

7.3 Basket position



Figure 7.3 — RFID tracking and the position of a selected basket

- 1 **RFID tracking** — pulse counter, basket at R0, loops and home checks (B001 must return after exactly 380 pulses).
- 2 **Position from R0** in metres and pitch number.
- 3 **Next station** with distance and time.
- 4 **To its unloader** — where the cone will be unloaded, distance and time.
- 5 **How it is calculated** — places ahead of the basket last read at R0 + pulses + encoder fraction, × 0.8 m, checked against the real chain position.
- 6 **Distance to every unloader.**

Screen guide — Master Mind panel

Clicking the brain opens the decision panel. It shows what the brain sees, how it ranks the loaders, the timing forecast, and the commands it sends — for the current second, or frozen.

8.1 Live thinking

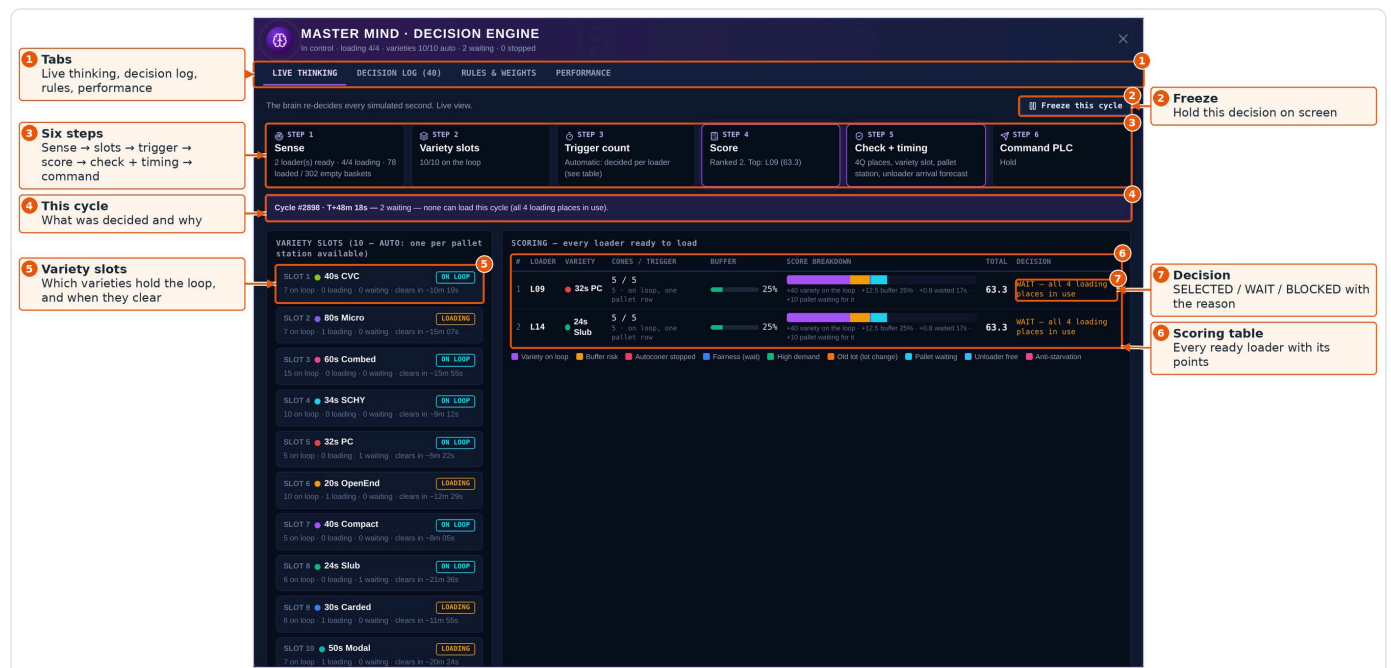


Figure 8.1 — Live thinking

- 1 Tabs** — Live thinking, Decision log, Rules & weights, Performance.
- 2 Freeze** — hold this decision on screen while the plant keeps running.
- 3 Six steps** — sense, variety slots, trigger count, score, check + timing, command PLC; highlighted when active.
- 4 This cycle** — what was decided and why, in one line.
- 5 Variety slots** — which varieties are on the loop (automatic: one slot per available pallet station), what each is doing and when it will have cleared. A red box explains any “drain”.
- 6 Scoring table** — every loader ready to load: cones and trigger count, buffer, score breakdown (coloured bar) and total.
- 7 Decision** — SELECTED with pallet station, unloader and pace, or WAIT / BLOCKED / HOLD with the reason. Selected loaders also show when their first and last cones reach the unloader.

8.2 Timing forecast and commands



Figure 8.2 — Timing card and commands to the PLC

- 1

Timing from the setup — basket every 4.8 s, loader cycle 24 s, unloader cycle 19.2 s, 1.25 streams per unloader, time until the next variety slot is free.
- 2

Unloader forecast — for each unloader: cones booked, next cone in, booked until, load and forecast returns.
- 3

Commands to PLC — each permit with its full reason: PERMIT_LOAD , LOAD_PACE , STATION .

8.3 Decision log

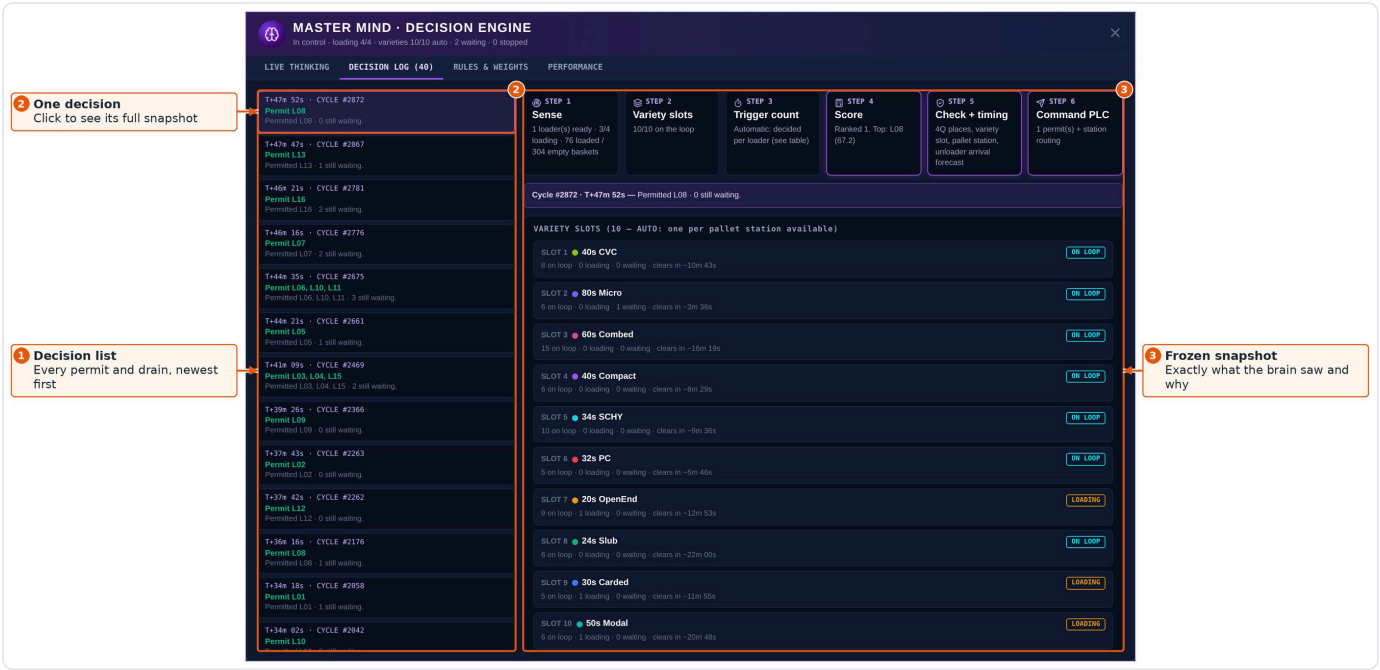


Figure 8.3 — Decision log

- 1

Decision list — every permit and drain, newest first.
- 2

One decision — click to open it.
- 3

Frozen snapshot — exactly what the brain saw and decided at that moment.

8.4 Rules and weights



Figure 8.4 — Hard rules, scoring weights and failure test

- 1 Hard rules** from the setup — never broken: loading places, variants on the loop, pallet station per lot, lots never mixed, start rule, loading count, timing, FIFO fallback.
- 2 Scoring weights** — sliders; the brain uses the new values immediately.
- 3 Drain timing** — how many minutes before a waiting loader would fill the brain starts freeing a variety slot. Below it are the pace and return-cost settings.
- 4 Defaults** — back to the tested weights.

“Simulate brain PC failure” at the bottom stops the brain: the plant continues in FIFO, as the PLC will do.

8.5 Performance

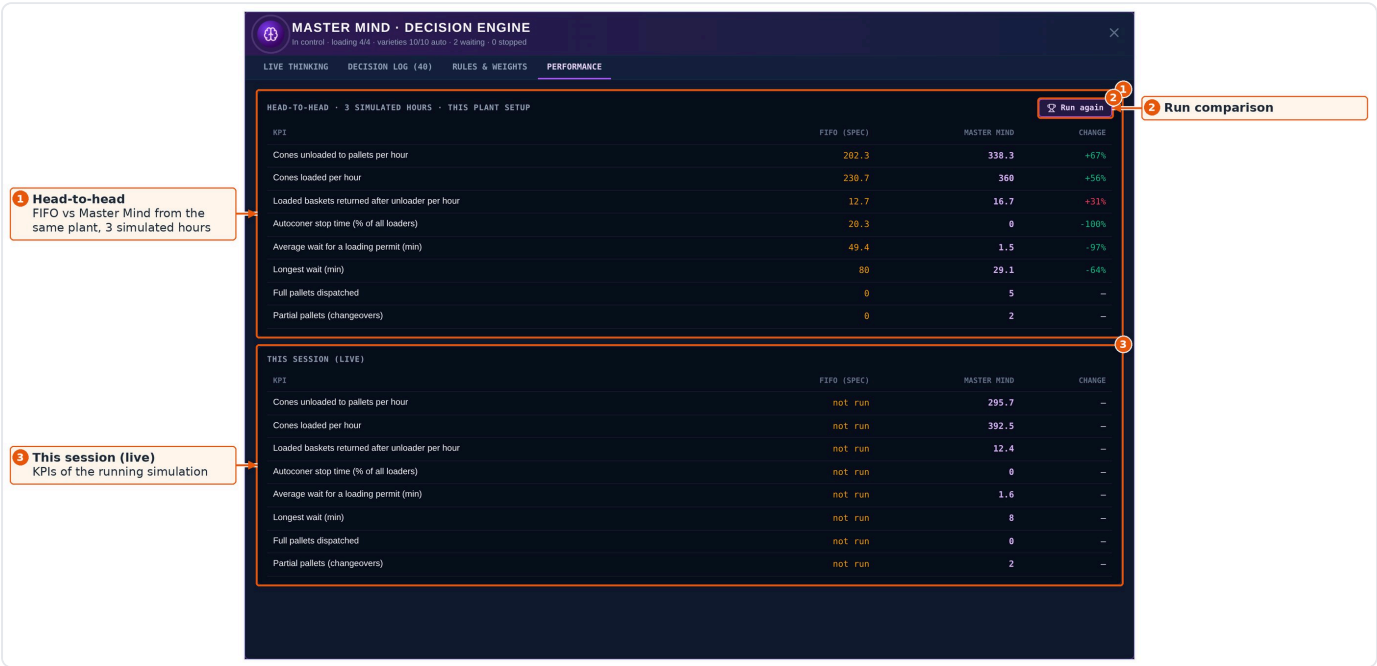


Figure 8.5 — Head-to-head comparison

- 1 Head-to-head** — FIFO and Master Mind from the same starting plant for 3 simulated hours: cones per hour, stop time, waiting, returns, pallets.
- 2 Run comparison** — runs both strategies in the background (a few seconds).
- 3 This session** — the KPIs of the simulation that is running.

How the Master Mind decides

Inputs

Source	Inputs
Plant setup (fixed)	Station positions from R0, pitch and speed (4.8 s), loading and unloading times (24 s / 19.2 s cycles), loader buffer, cones/h per variant, loader → variant, pallet stations per unloader, cones per accumulator and pallet, limits, rules and scoring weights.
PLC (every second)	Heartbeat, chain running, speed, pulse counter, basket at R0, encoder (→ every basket's position); per loader: cones waiting, old-lot cones, status, stopped, variety, lot; per unloader: ready/busy; per pallet station: status, variety, lot, accumulator and pallet count; per basket: empty/loaded/returned, variety, lot, target station.
Operator	Lot change, high-demand variants.

From these the brain works out: the time until each loader's buffer is full $((\text{buffer} - \text{cones}) \div \text{cones/h})$, how long each loader has waited, which variants are on the loop and when each will have left it, when every cone will reach its unloader (basket by basket), and which pallet stations are free.

The six steps (every second)



Step 2 — Variety slots

- **Limit.** Automatic mode: one variety per available pallet station (each variety/lot needs its own station). Fixed mode: the specification's 4.
- **Drain.** If a loader outside the loop cannot get a slot and its buffer will fill before any variety clears (with 15 min lead), the brain stops loading the variety that clears soonest and reserves its slot for that loader.

Step 3 — Trigger count (automatic loading count)

- Variety already on the loop: one pallet row (5 cones).
- Other varieties waiting for a slot: 85 % of the buffer, so the slot can rotate.
- New variety while slots are scarce: 50 % of the buffer — lowered if the cones arriving while it waits for a slot would not fit in the buffer.

Step 4 — Scoring

Reason	Points
Variety already on the loop / needs a new slot (only when slots are scarce)	+40 / -10
Buffer fill	up to +50
Autoconer already stopped	+30
Waiting time	+3 per min, max +30
High-demand variety (start rule includes demand)	+25
Old lot to clear after a lot change (high priority)	+100
Pallet already waiting for this variety (start rule includes unloader)	+10 (+20)
An unloader is free for it (start rule “unloader requirement”)	+15
Waiting more than 20 min (anti-starvation)	+500

Step 5 — Check and timing forecast

Going down the ranking, a loader is permitted only if: a loading place is free (4Q), its variety is not being drained, a variety slot is free or reserved for it, and a pallet station can take its lot. Then the **timing forecast** decides where and how fast:

- For every possible pallet station, the brain forecasts basket by basket when each of this loader’s cones reaches that station’s unloader, together with all cones already booked there. The unloader can take a cone only every 4th basket.
- It chooses the station where the batch is cleared soonest; every cone forecast to pass its unloader unloaded costs one loop (30 min 24 s).
- A loader is never sent cones faster than one unloader can take them (`LOAD_PACE`).
- If cones would still pass unloaded and the loader is not urgent, it waits.

Step 6 — Commands

`PERMIT_LOAD[Lxx] = 1` , `LOAD_PACE[Lxx] = N` (every Nth basket), `STATION[Lxx] = Pyy` , station assignment or changeover — or a permit is withdrawn (drain). Each decision is logged with its reason.

Example

L03 (30s Carded, variety on the loop, 7 of 20 cones, waited 2 min): $40 + 17.5 + 6 = 63.5$. **L08** (80s Micro, new variety, 16 of 20 cones, waited 6 min): $-10 + 40 + 18 = 48$; its buffer is full in about 7 min.

L03 is ranked first. Its variety’s station P04 is on U02 (160.8 m); from L03 (13.6 m) that is 147 m $\approx$ 14 min 43 s. U02 is booked for the next 13 min, so L03’s cones arrive after that and none return → *permit, every 5th basket, station P04*.

L08 cannot get a variety slot and would fill before any variety clears → the brain *drains* the variety that clears soonest and reserves that slot for L08.

Basket positions with RFID

Every basket carries an RFID tag. On a closed loop there is no natural start, so one fixed point is chosen as the **origin**: an RFID reader **R0** just before L01. R0 is 0 m and also the loop end (304 m).

Commissioning (once)

1. **Order list.** Run one full loop; R0 records the order of all 380 tags (B001, B002, ...). The order never changes — baskets are fixed to the chain.
2. **Station positions.** At one moment R0 reads basket X and a station's reader reads basket Y; the number of places between them in the order list $\times 0.8$ m is the station's distance from R0. (Or take them from the layout drawing.)
3. **Loop check.** The same tag must return to R0 after exactly 380 pitch pulses.

Formulas

What	Formula
Position of basket B (m from R0)	$((\text{place of the basket last read at R0} - \text{place of B}) \bmod 380 + \text{pulses since that read} + \text{encoder fraction}) \times 0.8$
Distance to a station	$(\text{station position} - \text{basket position}) \bmod 304 \text{ m}$
Time to a station	$\text{distance} \div 10 \text{ m/min}$
Home check	B001 read at R0 $\rightarrow$ pulse count must be a multiple of 380; otherwise a pulse was missed and positions are resynchronised

Worked example — R0 is reading B200

Basket	Places ahead	Position	To U01 (152.8 m)	Where
B150	50	40 m	112.8 m $\cdot$ 11 min 17 s	between L07 and L08
B010	190	152 m	0.8 m $\cdot$ 5 s	arriving at U01
B300	280	224 m	232.8 m $\cdot$ 23 min 17 s	past U04, on the empty return

In the simulation the same calculation is shown for any basket you click, with a check that it matches the real chain position (figure 7.3). Over three simulated hours, for four different plants, the RFID-based position matched the chain for every basket at every check.

PLC interface (Modbus TCP)

Register map (proposal)

All values are 16-bit holding registers, 0-based offsets (4xxxx = 40001 + offset). The block start addresses can be changed on the PLC page to match the PLC program. The full list (1,838 registers) is exported as CSV.

Block	Direction	Address	Per item
System	PLC → brain	0–15	PLC heartbeat, mode, chain running, speed ×10, pulse counter (2 words), basket at R0, encoder, fault word, home check, loops
Loaders (16 × 8)	PLC → brain	100–227	cones waiting, old-lot cones, status, autoconer stopped, variety, lot, loaded counter, basket in front
Unloaders (4 × 6)	PLC → brain	300–323	status, basket in front, unloaded counter, last cycle time
Pallet stations (11 × 8)	PLC → brain	400–487	status, variety, lot, accumulator, pallet, pallet present, pallets done
Baskets (380 × 4)	PLC → brain	1000–2519	state (empty/loaded/returned), variety, lot, target station
Brain	brain → PLC	3000–3001	brain heartbeat, brain active
Loader commands (16 × 4)	brain → PLC	3010–3073	permit, pace, station, trigger count
Station commands (11 × 4)	brain → PLC	3074–3117	variety, lot, command (assign / change over)

Gateway

- `npm run plc:gateway` starts it on the Master Mind PC (WebSocket ws://localhost:8765).
- The page sends PLC address, port, unit ID, poll time and the blocks to read; the gateway polls the PLC (at most 120 registers per request) and pushes the results.
- Writes are accepted only when “Allow the brain to write” is ticked. It reconnects automatically after a lost connection.

PLC simulator (testing without the PLC)

```
npm run plc:simulator (Modbus TCP on port 5020, customer defaults)
node plc-gateway/plc-simulator.js --setup sieger-setup.json --port 502
```

It answers Modbus exactly like the real PLC should, with random but consistent values, and prints every command the brain writes (for example “brain wrote L01.permit = 1 (address 3010)”). Tested end to end: page → gateway → Modbus → simulator, read and write.

To confirm at the site visit: PLC make and protocol, whether basket tracking is in the PLC, RFID readers at each station, the data layout, command handshake, and FIFO fallback when the brain heartbeat stops.

Results so far

Simulated, 4 hours each, same starting plant. Customer values from Requirements.xlsx plus the assumed values of chapter 13.

Customer setup

Strategy	Cones to pallets /h	Autoconer stop time	Returned baskets /h	Variants on loop
FIFO — customer specification (4 variants fixed)	204	22 %	22.5	4
Master Mind — 4 variants fixed	264	10 %	11.8	4
Master Mind — automatic variants	338	0 %	14.5	9–10

The autoconers produce 350 cones/h, so in automatic mode almost all of it is unloaded and no autoconer stops. In the 3-hour head-to-head of the app (figure 8.5) the result is the same: 202 vs 338 cones/h, stop time 20 % vs 0 %, average wait 49 vs 1.5 min. Returned baskets: in the 4-hour run the Master Mind returns fewer than FIFO (14.5 vs 22.5 /h); in the 3-hour head-to-head slightly more (16.7 vs 12.7 /h), because in automatic mode more varieties share the unloaders. This is the next point to improve.

Other plant variations

Plant	FIFO	Master Mind (4 fixed)	Master Mind (auto)
Unloading 30 s instead of 18 s	78 /h	211 /h	322 /h
8 pallet stations	212 /h	264 /h	334 /h
Loader buffer 30 cones	241 /h	285 /h	352 /h
6 variants × 60 cones/h	231 /h	311 /h	344 /h

Lot change

With two lot changes at six different times: FIFO clears the old lot in 81–128 min; the Master Mind (high priority) in 16–42 min, keeping each lot on its own pallet.

Checks

Checked every simulated second in every test: never more loaders loading than allowed, every loaded basket has a pallet station, accumulators and pallets never overfilled, loader cone counts always consistent, every lot change completed, RFID positions always correct. Lots never share a pallet by design (a station holds one lot).

Assumptions, open questions and next steps

Assumed values (not in the requirement sheet)

Value	Assumed	Effect
Loader buffer before the autoconer stops	20 cones	Stop risk, trigger count
Loader spacing / unloader spacing	6.4 m / 8 m	Travel times, positions
Origin R0 before L01	0.8 m	Position reference
Cones per pallet	100	Pallet changes
Loading and unloading on the move (chain never stops)	yes	Output: about 340 /h on the move, about 100 /h if the chain stops

Key questions for the site

1. Layout drawing with the position of every loader, unloader and the RFID origin reader.
2. Does the chain run continuously while loading and unloading, or stop at each station?
3. Loader buffer capacity before the autoconer stops.
4. Reason for the limit of 4 variants on the loop, and for the maximum of 4 loaders loading at once.
5. Pallet matrix: 4 or 5 cones per row per variant, cones per pallet, partial pallets.
6. Can pallet stations be reassigned automatically? Is the station → unloader assignment fixed?
7. PLC make, model and protocol (Modbus TCP, Profinet, OPC UA); existing data layout.
8. Basket tracking in the PLC: RFID reader at each station, pitch sensor or encoder.
9. How a lot change and “high demand” are informed (operator, ERP, autoconer).
10. Acceptance targets: throughput, zero autoconer stops, waiting time, returns.

The full list of 37 questions has been sent to the customer by e-mail.

Next steps

1. **Site visit** — watch loading, unloading and palletising; check PLC read and write with the customer's PLC team.
2. **Enter the confirmed values** in the setup and re-run the comparison.
3. **Agree the register map** with the PLC programmer (or adapt ours to their layout).
4. **Connect the brain to the live PLC data** — first in advisory mode (decisions shown, not written), then in control with FIFO fallback.
5. **Trial and acceptance** against the agreed targets.

Glossary

Term	Meaning
OHC	Overhead conveyor: a closed chain carrying 380 baskets, one cone each.
Pitch	Distance between two baskets (800 mm).
Autoconer	Winding machine producing cones; stops when its loader's buffer is full.
Loader / unloader	Station putting cones into / taking cones out of passing baskets.
Accumulator	Collects one row of 4 or 5 cones before it is placed on the pallet.
Variant / variety	A yarn type (count and material). Different variants never share a pallet.
Lot	Production batch of a variant; lots never share a pallet.
4Q	Maximum number of loaders loading at the same time (4).
FIFO	First in, first out: the customer's queue rule; also the PLC fallback.
Trigger count	Cones a loader collects before it asks to load.
Drain	Stopping new loading of a variety so it leaves the loop and frees a slot.
Returned basket	A loaded basket that passed all unloaders and goes round again (30 min 24 s).
R0	The RFID origin reader: 0 m and loop end; all positions are measured from it.
LOAD_PACE	Command: load every Nth basket (normally every 5th).
Modbus TCP	Industrial protocol used to read and write PLC registers over Ethernet.
Gateway	Program on the Master Mind PC that talks Modbus TCP to the PLC for the web app.